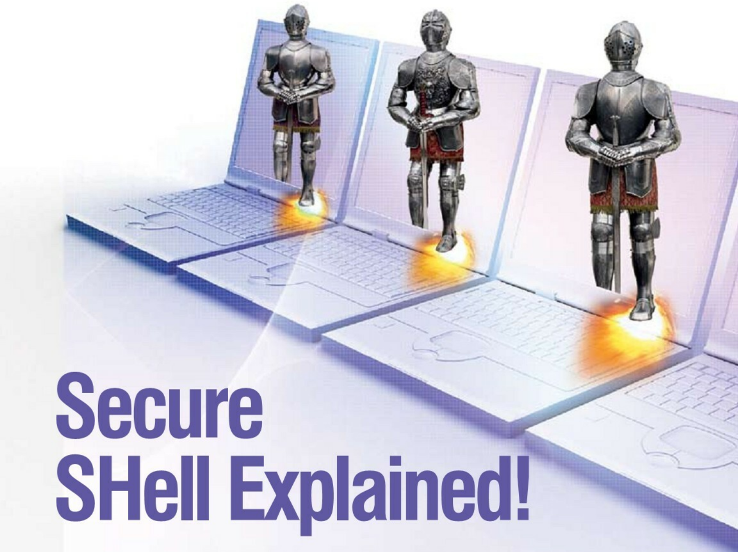

Secure SHell Explained!

Here're some insights into SSH (Secure Shell), an essential tool for accessing remote machines.

SSH is used to access or log in to a remote machine on the network, using its hostname or IP address. It's a secure network data exchange protocol that came up as a replacement for insecure protocols like *rsh*, *telnet*, *ftp*, etc. It encrypts the bi-directional data transfers using cryptographic algorithms, making the data transfers secure. Hence, it is free from password theft or from the sniffing of packets being transferred over a network.

Some of the highlights of the SSH protocol are:

- Compression
- Public key authentication
- Port forwarding
- Tunnelling
- X11 forwarding
- File transfer

SSH runs as a service daemon to facilitate remote log-ins. To install the SSH server on Debian-based distros, key in the following command:

```
# apt-get install openssh-server
```

Although the default port for SSH is 22, you can also configure it to run with other custom ports.

To perform remote log-ins, we require an SSH client. There are lots of SSH clients available, and they can be installed on Debian-based system as follows:

```
# apt-get install openssh-client.
```

It is possible to access remote UNIX/Linux machines from any other OSs using some SSH clients. For example, it is possible to remotely log in to a UNIX box from Windows using the SSH client called Putty [*www.putty.org*].